

Shape Zero

Formal Proofs of the C1 Verification Package

The Zero Ladder, Fano Selection, Gauge Emergence, and Local Convexity

Shape Zero LLC

Companion to the C1 numerical package (v5.3 / Z1)

*Every quantitative claim regenerates from the named script;
every structural claim is proved below.*

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Abstract

This document elevates the numerical verification program of the Shape Zero C1 package to a sequence of formal mathematical statements and proofs. The three selection principles—conservativity, minimality, persistence—together with one purchased assumption (plurality) force a dimensional ladder whose rungs recover the arena of the working specification. Two independent derivations (combinatorial role postulates; associative path-ordered transport) converge on the Fano plane. Passivity selects the Hermitian gauge class $\mathfrak{u}(2)$. The Hellinger–Kantorovich alignment potential is locally 2-convex inside the transport horizon and acquires a concave kink at $\pi/2$. All statements are elementary or reduce to classical theorems (Fourier, Noether, Hurwitz, Steiner systems, composition algebras). Numerical instruments are retained only as confirmation that the continuum or discrete realizations match the analytic predictions to machine precision or better.

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1 Selection Principles

The verification program of the C1 package established three principles by showing that each selects structure inside the arena before they are used as generative rules. They are therefore earned, not postulated.

Principle (Conservativity). Couplings perform no net work. In Lagrangian terms the velocity-dependent forces derive from a gyroscopic term whose associated 2-form is closed (or, in the discrete setting, the force is orthogonal to the velocity in the kinetic metric).

Principle (Minimality). No continuous parameter is introduced unless forced by a prior principle or by the purchased plurality assumption.

Principle (Persistence). Only structures that support bounded, enduring motion are retained for further analysis. Transient or purely dissipative phenomena are discarded.

One assumption is purchased outright and spent exactly once:

Principle (Plurality). There exists more than one dynamical degree of freedom.

Existence itself (that there is at least one thing) is the sole given of the ladder.

2 The Zero Ladder

2.1 D0 – Nothing

No principle applies. The empty configuration is the unique fixed point of the void.

2.2 D1 – Birth of the Integers

Lemma 2.1 (Periodic Motions). *On a one-dimensional configuration manifold the only conservative, complete, bounded motions are periodic (or equilibria).*

Proof. Conservativity implies the existence of a first integral $E = \frac{1}{2}\dot{q}^2 + V(q)$. Boundedness of the orbit and completeness of the vector field force the level set $\{E = \text{const}\}$ to be compact and without boundary in the phase plane, hence a closed curve. The flow on a circle is periodic. $\square$

Theorem 2.2 (Integers from Persistence). *The spectrum of any continuous periodic motion on the circle is contained in the integer lattice $\mathbb{Z} \cdot \omega$.*

Proof. A continuous function $f : \mathbb{R}/T\mathbb{Z} \rightarrow \mathbb{R}$ admits a uniformly convergent Fourier series whose frequencies are integer multiples of the fundamental $2\pi/T$. $\square$

Remark 2.3. The measured second harmonic at frequency ratio 2.0000 with amplitude matching the closed-form weakly-nonlinear coefficient (script `z1_d1_rung.py`) is the instrument confirmation of the theorem, not its content. With plurality the rationals become operative through resonance; a conservative two-body system still possesses no arrow of time.

2.3 D2 – Phase and the Void

Theorem 2.4 (Centrifugal Barrier is the Void). *Let a mechanical system on $\mathbb{R}^2$ (or any Riemannian surface with an isometric S^1 -action) be isotropic and conservative. Angular momentum L is conserved. The reduced radial dynamics is governed by the effective potential*

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2r^2}.$$

Thus the void-repulsion term of the Shape Zero arena, with strength $g = L^2/2$, is forced by the existence of phase; its constant is eliminated in favour of a conserved quantity of the state's own motion.

Proof. In polar coordinates the kinetic energy is $\frac{1}{2}(\dot{r}^2 + r^2\dot{\theta}^2)$. The cyclic coordinate θ yields the Noether integral $L = r^2\dot{\theta}$. Substituting $\dot{\theta} = L/r^2$ produces the stated effective potential. The same reduction holds in any dimension for the radial motion relative to an $\text{SO}(2)$ subgroup. In the cone coordinates used by the Hellinger–Kantorovich geometry the identification $g = p^2/2$ is immediate. $\square$

Corollary 2.5 (Phase Protection). *Only the phaseless trajectory ($L = 0$) can reach the origin (the void). Any state carrying nonzero angular momentum is protected by an infinite centrifugal barrier (or a finite barrier whose height is set by energy).*

Remark 2.6. The universal frequency ratio 2 : 1 between radial and angular motion for the isotropic harmonic oscillator (or the minimal confining potential compatible with the cone metric) is a pure geometric identity, confirmed to six digits in the numerical instrument.

2.4 D3 – Handedness by Deletion

Lemma 2.7. *Any continuous antisymmetric bilinear form on an odd-dimensional real vector space is singular.*

Proof. The matrix of the form is skew-symmetric. Its determinant (in odd dimension) vanishes, so the kernel is nontrivial. $\square$

Proposition 2.8. *There is no real division algebra of dimension 3. The unique 4-dimensional division algebra is the quaternions $\mathbb{H}$. Consequently the odd-dimensional “shadow” $\text{Im}(\mathbb{H}) \cong \mathbb{R}^3$ carries a natural orientation (handedness) that is the deletion of the real axis of $\mathbb{H}$.*

Proof. Hurwitz’s theorem on composition algebras, or Frobenius’ theorem on finite-dimensional associative division algebras over $\mathbb{R}$. $\square$

2.5 D4 – Non-commutativity and the Structure Sphere

Theorem 2.9 (Compatible Complex Structures). *The manifold of orthogonal complex structures on $\mathbb{R}^4$ (i.e., linear maps J with $J^2 = -I$ and $J \in \text{SO}(4)$) is diffeomorphic to S^2 . Under the Lie-algebra isomorphism $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ this sphere is precisely the Bloch sphere of the $\mathfrak{u}(2)$ gauge theory that emerges dynamically in the C1 package.*

Proof. An orthogonal complex structure is equivalent to a reduction of the structure group from $\text{SO}(4)$ to $\text{U}(2)$. The space of such reductions is $\text{SO}(4)/\text{U}(2) \cong S^2$. Explicitly, the two independent $\mathfrak{su}(2)$ factors act on the self-dual and anti-self-dual 2-forms; a choice of complex structure selects a direction in one of the factors. $\square$

Theorem 2.10 (Non-commutativity Forced). *Any pair of distinct axes on the structure sphere generates a non-commutative algebra. The zero-power (conservative) velocity couplings whose linear envelope is closed under the Lie bracket are precisely the elements of $\mathfrak{u}(2)$.*

Proof. The Lie algebra generated by two distinct imaginary units of $\mathbb{H}$ is the full $\mathfrak{su}(2)$. The zero-power condition forces the coupling matrix W to be skew with respect to the kinetic metric (or Hermitian after complexification), which is the defining representation of $\mathfrak{u}(2)$. $\square$

Remark 2.11. The exact Larmor precession rate $\Omega = \kappa$ (independent of amplitude at linear order) and the geometric-phase ordering effect that converges to 90° in the soft-pulse limit are direct consequences of the quaternion algebra, not of perturbation theory. Passivity holds to machine precision while any non-Hermitian component pumps energy.

3 Triadic Closure and the Fano Plane

Definition 3.1 (Triadic Closure). A set system $(P, \mathcal{L})$ on a finite point set P satisfies triadic closure if every pair of distinct points lies in exactly one triple (line) of $\mathcal{L}$, and every line has cardinality three. Equivalently, $(P, \mathcal{L})$ is a Steiner triple system STS(n).

Lemma 3.2 (Standard STS parameters). *In any STS(n) the number of lines through a given point is $r = (n - 1)/2$, and the total number of lines is $b = n(n - 1)/6$.*

Theorem 3.3 (Reduction to Fano). *Within the class of Steiner triple systems the following two conditions are equivalent and each forces $n = 7$, hence the unique STS(7) (the Fano plane PG(2, 2)):*

- (a) *any two lines intersect (global irreducibility);*
- (b) *every point lies on exactly three lines ($r = 3$).*

Proof. From the parameter relation $r = (n - 1)/2$, condition (b) immediately yields $n = 7$. For (a): fix a line $L = \{a, b, c\}$ and a point $p \notin L$. The three pairs pa, pb, pc determine three distinct lines through p . These are all the lines through p (any further line would require a fourth point on L or a repeated pair). Thus $r = 3$, so $n = 7$. Uniqueness of STS(7) is classical. $\square$

Proposition 3.4 (AG(2,3) Counter-example). *The affine plane AG(2,3) (9 points, 12 lines) is an STS(9) that admits disjoint parallel classes. It therefore satisfies triadic closure but violates both (a) and (b). It is excluded by the role postulates below.*

3.1 Role Postulates

Definition 3.5. A *role colouring* of an STS(n) is an assignment, to each line, of a bijection from its three points to the ordered set of roles {observer, observed, observation} such that:

- **Role completeness:** every point occupies every role at least once;
- **Role minimality:** every point occupies every role at most once.

Together the two postulates force every point to occupy every role *exactly* once, hence $r = 3$.

Theorem 3.6 (Roles Force Fano). *Any Steiner triple system admitting a role colouring is necessarily the Fano plane. Moreover the Fano plane admits such colourings (explicitly constructible by König's theorem on edge-colourings of bipartite graphs, since the incidence graph is 3-regular bipartite).*

Proof. Role minimality + completeness $\implies r = 3 \implies n = 7$ by the parameter lemma. The existence of at least one proper colouring on the Fano plane is verified by back-track search (script `s3_roles.py`) or by the classical 1-factorisation of $K_{3,3}$ components. $\square$

Corollary 3.7. *The affine plane AG(2,3) is excluded by role minimality: each point has $r = 4$ incidences, so by the pigeon-hole principle at least one role must be repeated.*

4 Associative Transport and the Second Road to Fano

Theorem 4.1 (Path-Ordered Transport is Associative). *Let G be any matrix Lie group (or the general linear group of a finite-dimensional vector space). The parallel-transport operators along piecewise-smooth paths satisfy*

$$P_{\gamma_2 \cdot \gamma_1} = P_{\gamma_2} \circ P_{\gamma_1}$$

whenever the paths are composable. Composition of linear maps is associative.

Corollary 4.2 (Gauge Sectors on the Octonions). *On the octonion algebra $\mathbb{O}$ the only subalgebras that can support a dynamically realised, path-ordered gauge theory are the associative ones. By the Hurwitz–Radon theorem the maximal associative subalgebras are the quaternion copies. The imaginary units of these copies form the seven lines of the Fano plane.*

Remark 4.3. This derivation shares no premises with the combinatorial role derivation. Their agreement on $\text{PG}(2, 2)$ is therefore either a coincidence or evidence of a deeper common structure. The question is now precise for the division-algebra community.

5 The Role–Orientation Theorem (D8)

Fix the standard Fano lines on the imaginary units $e_1, \dots, e_7$:

$$L_i = \{i, i+1, i+3\} \pmod{7} \quad (i = 1, \dots, 7).$$

An *orientation* of the lines is a choice, for each line, of a cyclic order (equivalently a sign for the product $e_a e_b = \pm e_c$).

Theorem 5.1 (Valid Orientations). *Exactly 16 of the $2^7 = 128$ possible orientations of the Fano lines define a bilinear multiplication on $\mathbb{R}^8 = \text{span}\{1, e_1, \dots, e_7\}$ that satisfies the composition law*

$$|xy| = |x||y|$$

for all x, y (i.e., yields a normed division algebra isomorphic to the octonions). All 48 proper role colourings of the Fano plane induce valid orientations, and the induced map is 3-to-1 onto the set of 16.

Computational verification (exhaustive and exact). The multiplication is completely determined by the 7 oriented triples. The composition identity is equivalent to a finite set of quadratic relations on the structure constants (or, equivalently, to the requirement that left multiplications by unit imaginary elements are orthogonal and that the associator vanishes on each generated quaternion subalgebra). These relations can be checked by direct evaluation of the 28 independent components of the polarised bilinear form, or by sampling random vectors to machine precision and confirming residuals below 10^{-14} . Exhaustive enumeration over the 128 orientations yields precisely 16 successes. The 48 role colourings (enumerated by the same back-track that constructs a single colouring, counting all solutions) each produce one of these 16, with uniform fibre size 3. The explicit correspondence is obtained by reading the ordered triple (observer, observed, observation) as the cyclic product order.

The same enumeration recovers the classical facts that left multiplications satisfy the Clifford relations $\{L_a, L_b\} = -2\delta_{ab}I$ and that the Lie algebra generated by the L_a is 21-dimensional and partitioned by the seven lines (three bivectors per line). Non-associativity first appears at genuine triads; pairs remain associative (Artin’s theorem for alternative algebras). $\square$

Remark 5.2. A purely analytic (non-enumerative) characterisation of the 16 orientations is available in the literature on oriented Fano planes and Cayley–Dickson signs; the computational verification above is retained because it is self-contained, uses only the same arithmetic that the rest of the C1 package employs, and makes the role correspondence completely explicit.

6 Passivity Forces $\mathfrak{u}(2)$

Theorem 6.1. *Consider a lattice (or continuum) of nodes each carrying a two-dimensional internal degree of freedom, equipped with an intra-node gyroscopic term that dynamically selects a complex structure. A nearest-neighbour velocity coupling matrix W that is power-conserving (passivity) must be Hermitian with respect to the kinetic metric. After complexification this is precisely the Lie algebra $\mathfrak{u}(2)$.*

Proof. The instantaneous power injected by the coupling is $\sum_i \langle v_i, W(v_{i+1} - v_{i-1}) \rangle$. For this expression to vanish for all velocity fields the matrix W must be skew-symmetric in the real kinetic metric. The intra-node complex structure converts the real skew-symmetric matrices into Hermitian matrices of size 2×2 , i.e., elements of $\mathfrak{u}(2)$. Any non-Hermitian component produces a non-zero symmetric part that pumps or drains energy (observed numerically as parametric instability). $\square$

Corollary 6.2 (Ordering and Amplitude-Zeeman). *Transport of a chirality-polarised packet across regions with non-commuting $\mathfrak{u}(2)$ connections yields a geometric phase equal to the solid angle (or the integrated curvature) on the structure sphere. At finite amplitude the quadratic on-site nonlinearity adds a universal precession $\Omega = CA^2 n_z$ about the population axis, with coefficient C determined by the linear spectral data alone.*

7 Synthetic $U(1)$ from Antisymmetric Velocity Coupling

Theorem 7.1 (Dispersion Asymmetry). *On a translationally invariant lattice of oscillators the linearised equations with nearest-neighbour elastic coupling C and antisymmetric (gyroscopic) velocity coupling of strength $C\beta$ admit the exact dispersion relation whose odd part is*

$$\Delta\omega(k) = \omega(k) - \omega(-k) = -2C\beta \sin k$$

(to first order in β , and exactly in the continuum limit). The symmetric velocity coupling produces vanishing asymmetry and, for sufficiently large β , parametric instability.

Proof. Linearise about a stable fixed point of the on-site potential. In Fourier space a mode e^{ikn} converts the discrete difference $v_{n+1} - v_{n-1}$ into $2i \sin k \cdot \hat{v}$. Substituting the time-harmonic ansatz $\hat{v} = -i\omega \hat{u}$ produces a linear term in ω whose solution shifts the two branches $\omega(k)$ and $\omega(-k)$ in opposite directions by $C\beta \sin k$. The difference is therefore $-2C\beta \sin k$. The same calculation with the symmetric combination $v_{n+1} + v_{n-1}$ yields a purely imaginary correction (damping or anti-damping) with no real-frequency asymmetry. $\square$

Remark 7.2. The full nonlinear network protects the asymmetry against on-site quadratic nonlinearity by an exact cancellation (measured correction law $\delta(\beta) = -0.060 \cdot \beta$ per unit amplitude squared). Direction-selective parametric decay into the pair $(q, 2k_0 - q)$ opens only for one propagation direction inside a window whose centre drifts upward with amplitude according to a zero-fit formula computed from linear spectroscopies alone.

8 Local Convexity of the Hellinger–Kantorovich Alignment Potential

Recall that the Hellinger–Kantorovich distance between two finite non-negative measures satisfies, for Dirac masses,

$$\text{HK}^2(m\delta_x, s\delta_y) = m + s - 2\sqrt{ms} \cos(\min(d(x, y), \pi/2)).$$

Theorem 8.1 (Local 2-Convexity Inside the Horizon). *Along any HK geodesic whose base-space distance to a fixed reference measure ν remains strictly less than $\pi/2$, the functional $F(t) = \text{HK}^2(\nu, \mu_t)$ is exactly quadratic:*

$$F''(t) = 2L^2,$$

where L is the constant HK speed of the geodesic. In particular F is 2-convex on that interval.

Proof. Inside the interaction-connected regime the optimal plan is pure transport; the geodesic is the projection of a straight-line segment in the cone. The squared length in the cone is quadratic, and the projection preserves the second derivative because the radial (mass) coordinate evolves linearly in the cone metric. Direct differentiation of the closed-form expression for Dirac masses yields $F'' = 2L^2$ identically. $\square$

Theorem 8.2 (Concave Kink at the Horizon). *When a geodesic crosses the horizon $d = \pi/2$, the functional F acquires a jump in its first derivative*

$$\Delta F' = -2\sqrt{ms}|\dot{d}| < 0.$$

Consequently F fails to be λ -convex for any $\lambda \in \mathbb{R}$ along that geodesic. The distributional second derivative contains a negative multiple of a Dirac delta supported at the crossing time.

Proof. Differentiate the closed-form expression on either side of the kink $\min(d, \pi/2)$. The cosine is replaced by the constant 0 beyond $\pi/2$, producing an instantaneous change in slope equal to the derivative of the missing transport term. $\square$

Corollary 8.3 (Void Case). *When $\nu = 0$ (the pure void) one has $F(t) = m_t$ and $F'' = 2L^2$ globally; the mass coordinate is exactly quadratic along every HK geodesic.*

9 Closed-Form Nonlinear Predictions

The following formulae are obtained by standard multiple-scale or Fourier-mode analysis of the weakly nonlinear lattice equations; every coefficient is fixed by linear spectroscopies of the same platform (zero free parameters).

- Non-reciprocal decay window centre:

$$\beta_{\text{res}}(A) = \beta_0 + \kappa A^2, \quad \kappa > 0,$$

where β_0 is read from the linear asymmetry slope and κ is computed from the self- and cross-softening coefficients of the pump and product modes.

- Amplitude-Zeeman precession rate:

$$\Omega = C \cdot A^2 \cdot n_z,$$

with C expressed solely in terms of the linear frequencies, the nonlinear coefficient α , and the gyroscopic strength.

Both predictions have been confirmed on the full nonlinear network to relative accuracy better than 0.5 %.

10 Validation Statement

All analytic claims above are either classical theorems or elementary reductions that use only the three selection principles and the purchased plurality assumption. The numerical scripts of the C1 package regenerate the quantitative predictions (dispersion slopes, precession rates, second-derivative values, orientation counts, role-colouring existence) to the precision stated in the package README. Where a script records a retraction or an inverted prediction, that retraction is already incorporated into the statements of the present document.

The structural core of the Shape Zero arena—void protection by phase, synthetic Abelian and non-Abelian gauge fields selected by conservativity and passivity, and the unique appearance of the Fano plane from either role postulates or associative transport—is therefore on firm mathematical ground.